INCH-POUND

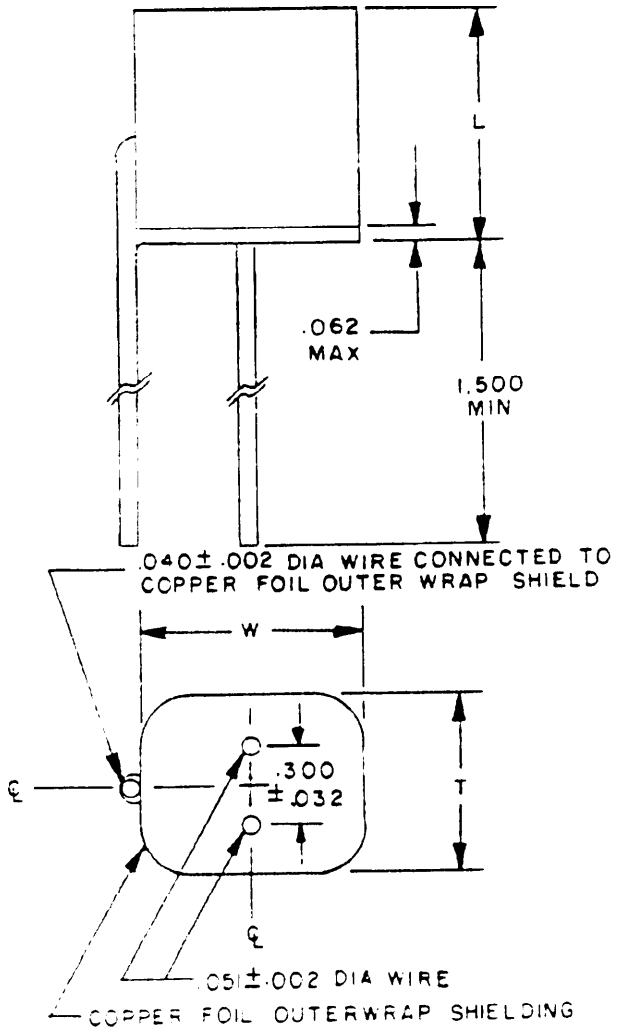
MIL-C-55514/12A(ER)
14 March 1990
SUPERSEDING
MIL-C-55514/12(ER)
2 June 1987

MILITARY SPECIFICATION SHEET

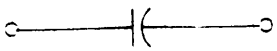
CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DC,
IN NONMETAL CASES, ESTABLISHED RELIABILITY
STYLE CFR29

This specification is approved for use by the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55514




STYLE CFR29



CIRCUIT DIAGRAM

FIGURE 1. Configuration and dimensions.



AMSC N/A
FSC 5910

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| Inches | mm |
|---|---|
| .002 | 0.05 |
| .040 | 1.02 |
| .051 | 1.30 |
| .062 | 1.57 |
| .093 | 2.36 |
| .156 | 3.96 |
| .187 | 4.75 |
| .300 | 7.62 |
| .437 | 11.10 |
| .468 | 11.89 |
| .500 | 12.70 |
| .562 | 14.27 |
| .593 | 15.06 |
| .718 | 18.24 |
| .750 | 19.05 |
| .781 | 19.84 |
| .862 | 21.89 |
| .875 | 22.23 |
| .906 | 23.01 |
| .937 | 23.80 |
| .968 | 24.59 |
| 1.000 | 25.40 |
| 1.031 | 26.19 |
| 1.093 | 27.76 |
| 1.250 | 31.76 |
| 1.281 | 32.54 |
| 1.312 | 33.32 |
| 1.375 | 34.93 |
| 1.437 | 36.50 |
| 1.500 | 38.10 |
| 1.687 | 42.85 |
| 1.812 | 46.02 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Configuration and dimensions - Continued.



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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case: Nonhermetic.

Dielectric: Metallized polycarbonate.

Characteristic: R.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +85°C, when properly voltage derated (see figure 2), characteristic "R" capacitors will operate to +125°C.

Voltage rating: See table I.

Preconditioning: Not applicable.

Burn-in: In accordance with MIL-C-55514 except for the following conditions:

a. DC burn-in voltage shall be 140 percent of the dc rated voltage applied for 48 hours minimum at 125°C +4°C, -0°C.

b. AC burn-in: Not applicable.

Insulation resistance: Insulation resistance shall be measured at rated voltage at the applicable temperature. Charge time 2 minutes maximum; however, for capacitance values greater than 1.0 µF, an additional 1 minute per µF is permitted. The following limits shall apply:

| Temperature | Insulation resistance |
|---|---|
| 25°C | 50,000 |
| 85°C | 5,000 |
| 125°C | 500 |

Marking: In accordance with MIL-C-55514.

Qualification submittal:

| Type designation | Sample size |
|---|---|
| CFR29RRA106FM | 61 |
| CFR29RRA506FM | 61 |
| CFR29RRG305FM | 61 |
| CFR29RRG506FM | 61 |
| CFR29RRB305FM | 61 |
| CFR29RRB506FM | 61 |



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TABLE I. Capacitor characteristics and dimensions.

| Type designation 1/ | Capacitance | Capacitance tolerance | Rated voltage at 85°C | ESR milliohms at 100 kHz | Ripple current max at 25°C | Resonant frequency kHz | Dimensions | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | T 2/ | W 2/ | L 3/ |
| CFR29RRA106-- | 10.0 µF | J K M | 50 V | 11 | 12.6 | 318 | .562 | .862 | .750 |
| CFR29RRA206-- | 20.0 µF | " " " | 50 V | 10 | 14.8 | 214 | .718 | .968 | .875 |
| CFR29RRA506-- | 50.0 µF | " " " | 50 V | 6 | 23.5 | 130 | 1.000 | 1.375 | 1.000 |
| CFR29RRG305-- | 3.0 µF | J K M | 75 V | 15 | 10.3 | 672 | .468 | .718 | .750 |
| CFR29RRG505-- | 5.0 µF | " " " | 75 V | 13 | 11.6 | 427 | .500 | .750 | .875 |
| CFR29RRG106-- | 10.0 µF | " " " | 75 V | 12 | 13.5 | 301 | .718 | 1.031 | .875 |
| CFR29RRG206-- | 20.0 µF | " " " | 75 V | 11 | 16.5 | 205 | .937 | 1.312 | 1.000 |
| CFR29RRG506-- | 50.0 µF | " " " | 75 V | 6 | 24.2 | 120 | 1.312 | 1.687 | 1.250 |
| CFR29RRB305-- | 3.0 µF | J K M | 100 V | 15 | 10.6 | 684 | .437 | .750 | .875 |
| CFR29RRB505-- | 5.0 µF | " " " | 100 V | 13 | 12.4 | 427 | .593 | .906 | .875 |
| CFR29RRB106-- | 10.0 µF | " " " | 100 V | 12 | 14.6 | 290 | .781 | 1.093 | 1.000 |
| CFR29RRB206-- | 20.0 µF | " " " | 100 V | 11 | 17.1 | 192 | .906 | 1.281 | 1.250 |
| CFR29RRB506-- | 50.0 µF | " " " | 100 V | 6 | 25.6 | 115 | 1.437 | 1.812 | 1.375 |

1/ The complete type designation will include additional symbols to indicate capacitance tolerance and failure rate level (M, P, R, and S).

2/ Tolerances: Thickness and width +.062, -.156 on .437 through .500 and +.093, -.187 on .562 and greater.

3/ Tolerances: Length +.093, -.062.



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Substitutability for failure rate levels shall be as follows:

| Failure rate level | Will replace failure rate level |
|---|---|
| S | R, P, M |
| R | P, M |
| P | M |
| M | |

Substitutability for capacitance tolerances shall be as follows:

| Capacitance tolerance | Will replace capacitance tolerance |
|---|---|
| F | G, J, K |
| G | J, K |
| J | K |
| K | |

Substitutability for voltages shall be as follows:

| Volts dc | Will replace voltage |
|---|---|
| 100 | 75, 50 |
| 75 | 50 |

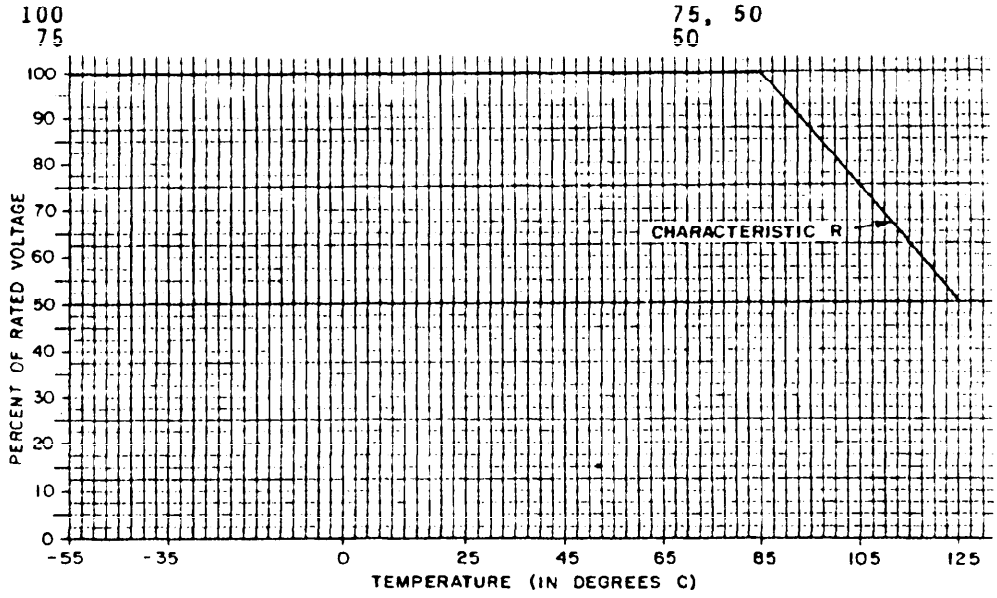



FIGURE 2. Voltage derating with temperature.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-A441)



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